Product Requirements Document (PRD)

Project Name

NetTrace

AI-Powered Network Forensics & Incident Reconstruction Platform

Tagline:

"Trace Every Packet. Reveal Every Attack."

Problem Statement

Modern organizations face an increasing number of cyberattacks, making network investigation more complex than ever. Security analysts often rely on multiple disconnected tools to analyze packet captures (PCAP files), investigate suspicious activities, reconstruct attack timelines, and prepare incident reports. This fragmented workflow consumes significant time, increases the possibility of missing critical evidence, and makes collaboration between security teams difficult.

Although tools like Wireshark provide detailed packet analysis, they primarily focus on raw packet inspection rather than complete incident investigation. Investigators must manually organize evidence, identify Indicators of Compromise (IOCs), document findings, and create reports using separate applications. This manual process slows down incident response and reduces operational efficiency.

There is a need for a centralized platform that simplifies the complete investigation process by combining packet analysis, case management, timeline reconstruction, IOC extraction, and report generation into a single, user-friendly application.

Objective

The primary objective of NetTrace is to simplify and streamline the network incident investigation process through a centralized platform.

The platform aims to:

- Provide a centralized environment for managing network investigation cases.
- Allow investigators to securely upload and analyze PCAP files.
- Automatically reconstruct attack timelines from captured network traffic.
- Detect and organize Indicators of Compromise (IOCs) such as suspicious IP addresses, domains, protocols, ports, and unusual network activities.
- Generate professional investigation reports for documentation and future reference.
- Reduce investigation time through an intuitive workflow and organized evidence management.
- Provide a beginner-friendly interface that can also support cybersecurity students learning network forensics.

Target Users

NetTrace is designed for individuals and organizations involved in cybersecurity investigations.

Primary target users include:

- Security Operations Center (SOC) Analysts
- Digital Forensics and Incident Response (DFIR) Teams
- Cybersecurity Professionals
- Security Researchers
- Ethical Hackers
- Cybersecurity Students
- Universities and Training Institutes
- Government Cybercrime Investigation Units
- CERT Teams
- Small and Medium Enterprises (SMEs)
- Large Organizations managing internal security operations

Features (V1)

The first version (V1) of NetTrace focuses on providing the essential features required for efficient network incident investigation.

Authentication

- Secure user login
- User registration
- Password reset

Dashboard

- Investigation overview
- Recent cases
- Quick statistics

Case Management

- Create investigation cases
- Edit case information
- Track investigation status
- View completed investigations

PCAP Upload

- Upload PCAP files
- File validation
- Secure storage

Packet Analysis

- Basic packet inspection
- Protocol identification

- Source and destination analysis
- Traffic summary

Incident Timeline

- Automatic timeline generation
- Chronological event visualization

IOC Detection

- Suspicious IP addresses
- Domains
- Ports
- Protocols
- Network endpoints

Report Generation

- Investigation summary
- Timeline summary
- IOC summary
- Exportable investigation report

Search & Filter

- Search investigation cases
- Filter uploaded evidence
- Filter timeline events

Responsive Interface

- Desktop support
- Tablet support
- Mobile support

User Flow

The user journey is designed to provide a simple and structured investigation process.

```
User Login
    |
    ▼
Dashboard
    |
    ▼
Create Investigation Case
    |
    ▼
Upload PCAP File
    |
    ▼
Packet Analysis
    |
    ▼
IOC Detection
    |
    ▼
Incident Timeline Generation
    |
    ▼
Review Investigation
    |
    ▼
Generate Investigation Report
```

|

▼

Save Case

|

▼

View Previous Investigations

Success Metrics

The success of NetTrace will be evaluated using the following metrics:

User Experience

- Easy-to-use interface
- Smooth navigation
- Responsive design across devices

Investigation Efficiency

- Reduced investigation time
- Faster evidence organization
- Improved workflow

System Performance

- Fast PCAP upload
- Quick packet analysis
- Stable application performance

Accuracy

- Correct timeline generation
- Accurate IOC extraction
- Reliable report generation

User Satisfaction

- Positive feedback from users
- Improved learning experience for students
- Better productivity for investigators

Future Scope

The future versions of NetTrace will expand the platform with advanced cybersecurity capabilities.

AI-Powered Investigation Assistant

- AI-generated incident summaries
- Investigation recommendations
- Automated anomaly detection

Live Network Monitoring

- Real-time packet capture
- Live traffic visualization
- Continuous monitoring

Threat Intelligence Integration

- VirusTotal Integration
- AbuseIPDB Integration
- AlienVault OTX Integration
- Threat Intelligence Feeds

MITRE ATT&CK Mapping

- Automatic attack technique mapping
- Threat actor behavior visualization

Collaboration Features

- Team-based investigations
- Shared workspaces
- Role-Based Access Control (RBAC)

Advanced Reporting

- Interactive dashboards
- PDF reports

- CSV export
- Executive summary reports

Enterprise Features

- Multi-organization support
- SIEM integration
- Audit logging
- API integrations

Cloud Deployment

- Multi-cloud support
- Scalable architecture
- Enterprise security enhancements

Conclusion

NetTrace is designed to provide a centralized and efficient platform for network forensics and incident reconstruction. By integrating packet analysis, investigation case management, IOC detection, timeline reconstruction, and report generation into a single application, the platform aims to reduce investigation complexity and improve overall cybersecurity incident response.

The proposed V1 focuses on delivering the core functionality required for effective network investigations while maintaining a scalable architecture for future enhancements such as AI-assisted analysis, live network monitoring, threat intelligence integration, and enterprise-level collaboration features.